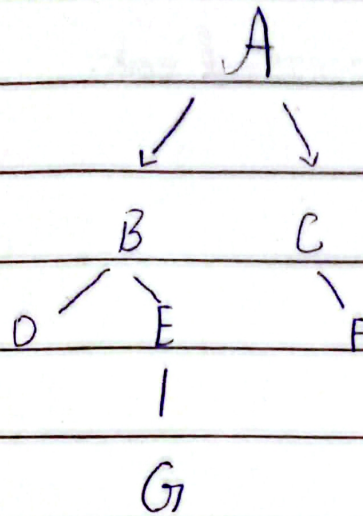


Graph:-



BFS Trace:

Queue	Path
[A]	A
[B, C]	A → B
[C, D, E]	A → C
[D, E, F]	A → B → D
[E, F]	A → B → E
[F, G]	A → C → F
[G]	A → B → E → G

Processed node: A, B, C, D, E, F, G

Final path A → B → E → G

DAY: _____

DATE: _____

DFS Trace:

Stack	processed node	Path
[A]	A	A
[C, B]	B	A → B
[C, E, D]	D	A → B → D
[C, E]	E	A → B → E
[C, G]	G	A → B → E → G

DFS visit order: A → B → D → E → G

Final path: A → B → E → G

Verification:

The BFS And Dfs
Traces were verified against the
given code and both
algorithm produce the final
path A → B → E → G.

DAY: _____

DATE: _____

2. DLS Result:

$A \rightarrow B \rightarrow E \rightarrow G$

depth counting:

$A = 0$

$B = 1$

$E = 2$

$G = 3$

DLS limit	Status	Path
1	Cut off	—
2	cut off	—
3	found	$A \rightarrow B \rightarrow E \rightarrow G$
4	found	$A \rightarrow B \rightarrow E \rightarrow G$

3. IDDFS Results:

Maximum depth	status	path	limits
1	failure	—	0, 1
2	failure	—	0, 1, 2
3	found	$A \rightarrow B \rightarrow E \rightarrow G$	0, 1, 2, 3
4	found	$A \rightarrow B \rightarrow E \rightarrow G$	0, 1, 2, 3

Important: In IDDFS, once the goal is found at depth 3, it stops. Therefore when maximum depth is 4, limit 4 is not

DAY: _____

DATE: _____

actually searched

4. Why limits below 3 fails:

The goal G is at depth 3.

Limit 1: Search can only reach $A \rightarrow B$ or $A \rightarrow C$.

Limit 2: Search can reach E , but not its child G .

Limit 3: Search can reach G , so the goal is found.

Therefore:

Limits 1 and 2 cannot succeed because they do not allow the search to reach the goal at depth 3.

5. Repeated work in IDDFS:

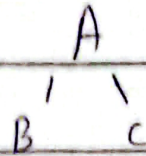
IDDFS repeatedly searches the upper part of the tree:

DAY: _____

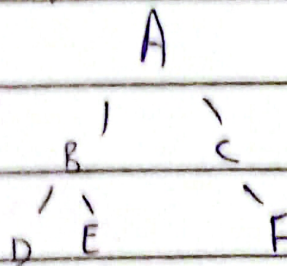
DATE: _____

limit 0: A

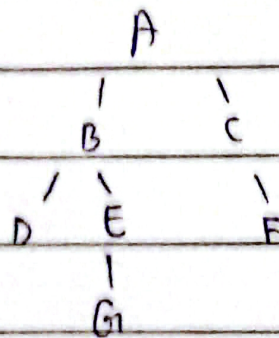
limit 1:



limit 2



limit 3



So nodes A, B, C, D and E may be
examine again at successive depth limit.

This repeated work is the main

disadvantage of IDDFS. However it

allows IDDFS to combine the low memory
usage of DFS with the shallowest solution

property of BFS when the step cost are equal.